UNIT 3 · STANDARD 6.RP.3

Compare Ratios

Lesson 3-5

Name: ______________ Date: ______________ Class: ______________

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

\$3 per 1 pound → unit rate is \$3/lb

Unit rate

A rate for just 1 of something, like cost for 1 item.

4:6 and 2:3 are equivalent (both simplify to 2:3)

Equivalent ratio

Two ratios that mean the same thing.

3:4 vs. 2:5 — which has more of the first ingredient per unit?

Compare

To look at ratios and see which is bigger, smaller, or equal.

8:12 → divide both by 4 → 2:3

Simplify

To make a ratio as small as possible while keeping the same comparison.

To compare 3/5 and 4/7, use LCD 35: 21/35 vs 20/35

Common denominator

A bottom number that two fractions can share so you can compare them.

Key Ideas & Notes

- Chef Academy is holding a Flavor Challenge!
- Chef Reyes's hot cocoa recipe uses 3 tablespoons of cocoa for every 5 ounces of milk.
- Chef Tran's recipe uses 4 tablespoons of cocoa for every 7 ounces of milk.
- The students need to figure out whose hot cocoa is more chocolatey.
- Build equivalent ratios for each chef's recipe so you can compare them fairly. Chef Reyes: 3 tbsp cocoa per 5 oz milk. Chef Tran: 4 tbsp cocoa per 7 oz milk.

Think About It

- What quantities are being compared in each recipe?
- Can you tell just by looking which recipe has more cocoa per ounce?
- What would make the comparison fair?

My Notes

Turn & Talk — Discussion Points

Level 1 support   Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

Chef Reyes uses 3 tbsp cocoa for every 5 oz of milk, and Chef Tran uses 4 tbsp for every 7 oz. Can you tell whose cocoa is more chocolatey just by looking at the numbers? Why or why not?

Level 1 support

Start here: You can't compare fairly until the ratios share the same amount of milk or you find cocoa per 1 oz.

You ____ tell just by looking because the ____ amounts are different.
____ se puede saber solo mirando porque las cantidades de ____ son diferentes.

To compare fairly we need the same ____ or the cocoa per ____.
Para comparar de forma justa necesitamos la misma ____ o el cacao por ____.

Word bank: ratio | compare | unit rate | equivalent ratio | fair

Listen for: *Students recognize that 3:5 and 4:7 have different milk amounts, so a fair comparison needs equal milk (a common amount) or a unit rate (cocoa per 1 oz).*

Level 2

Some students guess Chef Tran is more chocolatey because 4 is more than 3. Why is that reasoning not enough? Justify using the milk amounts.

- Looking only at the cocoa is not enough because ____.
- More cocoa does not always mean more chocolatey since ____.

EXPLORE

You scaled both recipes to 35 oz of milk: Chef Reyes became 21:35 and Chef Tran became 20:35. How does scaling to a common denominator let you compare the recipes fairly?

Level 1 support

Start here: When both ratios share 35 oz of milk, you only have to compare the cocoa amounts.

After scaling to 35 oz, Chef Reyes has ___ tbsp and Chef Tran has ___ tbsp.
Después de escalar a 35 oz, Chef Reyes tiene ___ cucharadas y Chef Tran tiene ___ cucharadas.

Now I can compare because the ___ is the same, so I just compare the ___.
Ahora puedo comparar porque la ___ es igual, así que solo comparo el ___.

Word bank: **equivalent ratio** **common denominator** **compare** **simplify** **milk**

Listen for: *Students explain that equal milk (35 oz) makes the comparison fair, so 21 tbsp > 20 tbsp means Chef Reyes is more chocolatey.*

Level 2

Find the unit rate (cocoa per 1 oz) for each chef and show it gives the same winner as the 35 oz scaling. Why must both methods agree?

- The unit rates are about ___ and ___ tbsp per oz, so ___ wins.
- Both methods agree because ___.

CONNECT

Two pizza shops have deals: Mario's sells 2 large pizzas for $18, and Sal's sells 3 for $25. How would you compare these to find the best price per pizza?

Level 1 support

Start here: Find the unit rate (price per 1 pizza) for each shop, then compare those rates.

Mario's costs ___ per pizza because 18 divided by 2 equals ___.

Mario's cuesta ___ por pizza porque 18 entre 2 es ___.

The better deal is ___ because its ___ is lower.

La mejor oferta es ___ porque su ___ es más baja.

Word bank: **unit rate** **compare** **ratio** **better deal** **per**

Listen for: *Students compute $9.00/pizza for Mario's and about $8.33/pizza for Sal's, then choose Sal's because the lower unit rate is the better deal.*

Level 2

If you only need 2 pizzas, does the better unit rate always mean the better total price? Explain when buying the deal with the lower unit rate might not save money.

- A lower unit rate is the best total deal only when ___.
- It might not save money if ___.

Guided Examples

Example 1

Chef A uses 2 cups of cheese for every 8 crackers. Chef B uses 3 cups of cheese for every 9 crackers. Who uses more cheese per cracker?

Solution: Chef A: $2 \div 8 = 0.25$ cups per cracker. Chef B: $3 \div 9 \approx 0.33$ cups per cracker. Chef B uses more cheese per cracker.

Answer: A. Chef B

Example 2

Which ratio is greater: 3:4 or 5:8?

Solution: Convert to the same denominator: 3:4 = 6:8. Since 6:8 > 5:8, the ratio 3:4 is greater.

Answer: A. 3:4

Example 3

Which lemonade recipe is more lemony? Recipe A: 2 lemons for 6 cups of water. Recipe B: 3 lemons for 7 cups of water.

Solution: Recipe A: $2 \div 6 \approx 0.33$ lemons per cup. Recipe B: $3 \div 7 \approx 0.43$ lemons per cup. Since $0.43 > 0.33$, Recipe B is more lemony.

Answer: A. Recipe B

Write About the Math

The Writing Revolution

I can explain my comparison using the words unit rate, equivalent ratio, compare, and common denominator.

1. Kernel Sentence

subject + verb

Model: Compare is to look at ratios and see which is bigger, smaller, or equal.
Comparar es mirar razones para ver cuál es mayor, menor o igual.

Write a kernel sentence about compare. Use a subject and a verb.
Escribe una oración base sobre comparar. Usa un sujeto y un verbo.

____

____

2. Sentence Expansion

because · but · so

Kernel: Compare matters in math
Comparar importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because *porque*

Compare matters in math because ____.
Comparar importa en matemáticas porque ____.

____

but *pero*

Compare matters in math, but ____.
Comparar importa en matemáticas, pero ____.

____

so *entonces*

Compare matters in math, so ____.
Comparar importa en matemáticas, entonces ____.

____

3. Sentence Types

4 ways to write a math idea

Statement
Afirmación

Tell one true fact about compare.

Di un hecho verdadero sobre compare.

Compare ____.

Question
Pregunta

Ask a question about compare.

Haz una pregunta sobre compare.

How does ____ ?

¿Cómo ____ ?

Exclamation
Exclamación

Show excitement about compare.

Muestra entusiasmo sobre compare.

Wow, ____ !

¡Guau, ____ !

Command
Mandato

Tell a partner what to do with compare.

Dile a un compañero qué hacer con compare.

First, ____ .

Primero, ____ .

4. Explain Your Reasoning

use a sentence starter

You ____ tell just by looking because the ____ amounts are different.

____ se puede saber solo mirando porque las cantidades de ____ son diferentes.

To compare fairly we need the same ____ or the cocoa per ____.

Para comparar de forma justa necesitamos la misma ____ o el cacao por ____.

After scaling to 35 oz, Chef Reyes has ____ tbsp and Chef Tran has ____ tbsp.

Después de escalar a 35 oz, Chef Reyes tiene ____ cucharadas y Chef Tran tiene ____ cucharadas.

Try It

Solve on your own. Check the answer key when you are done.

1. Store A sells 3 mangoes for $6. Store B sells 5 mangoes for $9. Which store offers a better price per mango?

A. Store B

B. Store A

C. Same price

D. Cannot determine

Show your work:

2. Store A sells 5 notebooks for $12. Store B sells 8 notebooks for $20. Without calculating, a student says Store B must be cheaper because you get more notebooks. Is the student's reasoning valid? Find the unit rates to prove your answer.

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

Find Kai's Mistake — find the error, then write the correct reasoning.

Show your work:

Reflect — Exit Ticket

A bakery sells 4 muffins for $10 and another bakery sells 6 muffins for $14. Which bakery has the lower price per muffin?

A. The second bakery
B. The first bakery
C. Same price
D. Cannot determine

Your answer:

Answer Key & Teacher Guide

1. **Try It 1:** A. Store B *— Store A: $6÷3 = $2.00 per mango. Store B: $9÷5 = $1.80 per mango. Store B has the better (lower) price per mango.*
2. **Try It 2:** The student's reasoning is not valid — getting more items does not mean a lower price per item. Store A: $12 ÷ 5 = $2.40 per notebook. Store B: $20 ÷ 8 = $2.50 per notebook. Store A is actually cheaper per notebook, even though Store B offers more notebooks per purchase.
3. **Exit Ticket:** A. The second bakery *— First bakery: $10÷4 = $2.50 per muffin. Second bakery: $14÷6 ≈ $2.33 per muffin. The second bakery has the lower price.*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Compare is to look at ratios and see which is bigger, smaller, or equal.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).